Library Management System

This is a simple library management system built in Python. It allows users to manage book records in a library and perform the following operations

- **Add books**: Add new books by entering their title and author.
- **View available books**: Display a list of all books with their current availability status.
- **Borrow books**: Borrow books by entering their unique IDs.
- **Return books**: Return borrowed books using their unique IDs.
- **Exit the system**: Exit the application

1. Adding Books

```
def add_book(title, author):
    book_id = random.randint(1000, 9999)  # Generate random 4-digit ID
    book = {"id": book_id, "title": title, "author": author, "available": True}  # Book dictionary
    library_books.append(book)  # Add book to list
    print("Book '" + title + "' added with ID " + str(book_id) + ".")
```

- This function adds a new book to the library_books list.
- A random ID (1000-9999) is assigned to each book.
- A dictionary with id, title, author, and available status is created.

2. Viewing Books

```
def view_books():
    if not library_books:
        print("No books in the library.")
        return

    print("\n--- Available Books ---")
    for book in library_books:
        status = "Available" if book["available"] else "Not Available"
        print(str(book["id"]) + " - " + book["title"] + " by " + book["author"] + " [" + status + "]")
```

- Displays the list of books along with their ID and availability.
- If no books are present, it prints "No books in the library.".

3. Borrowing Books

```
def borrow_book(book_id):
    for book in library_books:
        if book["id"] == book_id:  # Find the book by ID
            if book["available"]:  # Check if available
                book["available"] = False  # Mark as borrowed
                print("You have borrowed '" + book["title"] + "'.")
            else:
                print("Sorry, this book is currently not available.")
            return
    print("Book ID not found.")
```

- Searches for the book with the given book_id.
- If available, it changes available to False (borrowed).
- If not available, it prints "Sorry, this book is currently not available.".

- If no book is found, it prints "Book ID not found.".

4. Returning Books

```
def return_book(book_id):
    for book in library_books:
        if book["id"] == book_id:  # Find book by ID
            book["available"] = True  # Mark as available
            print("Thank you for returning '" + book["title"] + "'.")
            return
    print("Book ID not found.")
```

- Finds the book by ID and marks it as available (True).
- If the book is not found, it prints "Book ID not found.".

5. Implementing the Menu System

```
while True:
    print("\nLibrary Management Menu")
    print("1. Add Book")
    print("2. View Books")
    print("3. Borrow Book")
    print("4. Return Book")
    print("5. Exit")

    choice = input("Enter your choice: ")
```

· Infinite loop (while True) keeps running until the user chooses to exit.

6. Handling User Input

a) Adding a Book

```
if choice == '1':
    title = input("Enter book title: ")
    author = input("Enter book author: ")
    add_book(title, author)
```

- **Prompts** the user to enter book title & author.
- Calls add_book(title, author) to store the book.

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b) Viewing Books

```
elif choice == '2': view_books()
```

- Calls view_books() to **display all books**.

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c) Borrowing a Book

```
elif choice == '3':
    try:
        book_id = int(input("Enter Book ID to borrow: "))
        borrow_book(book_id)
    except ValueError:
        print("Invalid input. Please enter a valid number.")
```

- **Asks for Book ID** and converts it to an integer.
- Calls borrow_book(book_id).
- Handles **invalid input** (non-numeric values).

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d) Returning a Book

```
elif choice == '4':
    try:
        book_id = int(input("Enter Book ID to return: "))
        return_book(book_id)
    except ValueError:
        print("Invalid input. Please enter a valid number.")
```

- Asks for **Book ID** and calls return_book(book_id).
- Handles **invalid input**.

---

e) Exiting the Program

```
elif choice == '5': print("Exiting Library Management System.") break
```

- Ends the loop when the user chooses **"Exit"** (choice == '5').

Library used in project:-

import random:- for generates a random book ID for each new book.

```
import random
```

You can run this program in any compiler like python idle ,vs code PyCharm etc